

Vijay Makwana

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EDUCATION

IIT Kanpur

B.Tech in Civil Engineering

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Kanpur, Uttar Pradesh

- **Relevant Coursework:** Fundamentals of Computing, Probability and Statistics, Linear Algebra, Differential Equations, Introduction to Machine Learning, Computer Networks

PROJECTS

Agentic Recipe AI System | Python, Qdrant, Langgraph, HuggingFace, Tavily, Groq

- Designed and implemented an **agentic RAG pipeline using LangGraph**, enabling multi-step reasoning with tool-calling (ReAct architecture).
- Built **semantic retrieval system using Qdrant vector database and BAAI/bge-m3 embeddings**, improving context relevance and search accuracy.
- Integrated **Groq LLM APIs with dynamic tool orchestration**, enabling fallback from vector search to web search using **Tavily** for robust query handling.
- Engineered **context grounding and prompt pipelines**, reducing hallucinations and improving response quality in generated recipes.

RAG Document Assistant | Python, Pinecone, Groq

- Designed and implemented an end-to-end **Retrieval-Augmented Generation (RAG)** pipeline integrating LLMs with document retrieval for context-aware question answering.
- Built semantic search using **FAISS and Sentence Transformers**, enabling efficient vector indexing and high-accuracy document retrieval.
- Integrated LLM inference via Groq APIs and engineered prompt pipelines to improve response grounding and reduce hallucinations.

LoRA Finetuning | Python, PEFT, Tensorflow

- Fine-tuned a **1.5B parameter LLM** using **LoRA (PEFT)** on a single T4 GPU.
- Applied **4-bit quantization and memory-efficient training** to enable low-resource model adaptation.
- Built scalable training pipeline with checkpointing and gradient accumulation.
- Gained hands-on expertise in production-ready LLM customization.

Dialogue Summarizer - LLM Fine-Tuning | Python, Tensorflow, Hugging Face

- Fine-tuned **FLAN-T5** for multi-turn conversational summarization.
- Optimized training under compute constraints and improved inference latency.
- Deployed model on **Hugging Face Spaces** with interactive Gradio UI.

Detect Duplicate Questions - Machine Learning | Python, NLP, sci-kit learn

- Built semantic similarity classifier using TF-IDF, embeddings, and ML models.
- Benchmarked Logistic Regression and XGBoost with cross-validation.
- Engineered scalable preprocessing pipeline for large datasets.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), C/C++, SQL

GenAI & NLP: LLM Fine-Tuning, LoRA (PEFT), RAG, Prompt Engineering, Agentic AI, Transformer Models

Frameworks: PyTorch, Keras, Hugging Face Transformers, LangChain, FAISS, Sentence Transformers, Scikit-learn

Core CS: DSA, OOP, DBMS, Computer Networks

Soft Skills: Time Management, Leadership, Problem Solving, Team Working, Adaptability, Analytical Thinking, Collaborations